



ASSIGNMENT - 09

NLP Basics

DATA SCIENCE

NATURAL LANGUAGE PROCESSING - BASICS

1. What is Natural Language Processing (NLP)?
2. Name three common applications or tasks of NLP.
3. What is tokenization in NLP?
4. What are stop words, and why might we remove them from text?
5. What is the difference between stemming and lemmatization?
6. What is part-of-speech tagging?
7. What is Named Entity Recognition (NER)?
8. What is the Bag-of-Words (BoW) model in text processing?
9. What is TF-IDF and why is it useful?
10. What is an n-gram in the context of NLP?
11. What is sentiment analysis?
12. What is the difference between syntax and semantics in language processing?
13. What are word embeddings?
14. What does “out-of-vocabulary (OOV)” mean in NLP?
15. What are some typical preprocessing or text normalization steps for raw text data?
16. How does a rule-based NLP system differ from an ML/DL-based NLP system?
17. What is a language model in NLP?
18. What metrics could you use to evaluate the performance of a text classification model (for example, an email spam classifier)?
19. How are regular expressions used in text processing? Give an example.
20. What is a text corpus?
21. What is one-hot encoding in the context of NLP, and what is a limitation of one-hot encoded word vectors?
22. What are unigrams, bigrams, trigrams, etc.?
23. Name a popular Python library for NLP and something it's commonly used for.
24. What is the difference between a vocabulary and a corpus in NLP?
25. How can slang, misspellings, or abbreviations in text be handled during preprocessing?
26. What are two levels at which text can be tokenized? (Give examples.)
27. What are some challenges in NLP caused by language ambiguity or context variability?
28. Name two popular pre-trained word embedding models.
29. What is the “bag-of-words assumption” and what is one limitation of it?
30. Why is context important when interpreting the meaning of a word or sentence?